

MiniQuiz 5.2

Consider the initial-value problem on $1 \leq t \leq 2$ with initial condition $y(1) = -1$ and exact solution $y(t) = -\frac{1}{t}$:

$$y' = \frac{1}{t^2} - \frac{y}{t} - y^2, \quad 1 \leq t \leq 2, \quad y(1) = -1,$$

- (a) Use Euler's method with step size $h = 0.05$ to approximate the solution on $[1, 2]$, and compare the numerical approximations with the exact values $y(t) = -\frac{1}{t}$.
- (b) Using the Euler values computed in part (a) and linear interpolation, approximate the values

$$\text{i. } y(1.052), \quad \text{ii. } y(1.555), \quad \text{iii. } y(1.978),$$

and compare these interpolated approximations with the corresponding exact values.